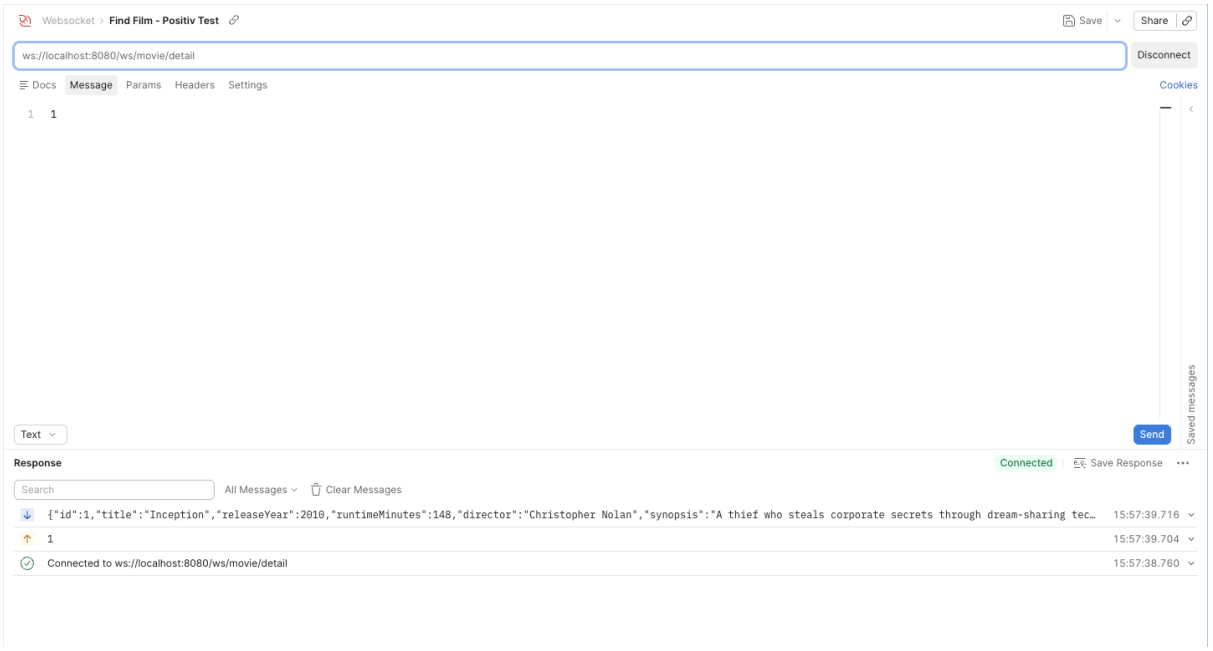
Websockets Tests

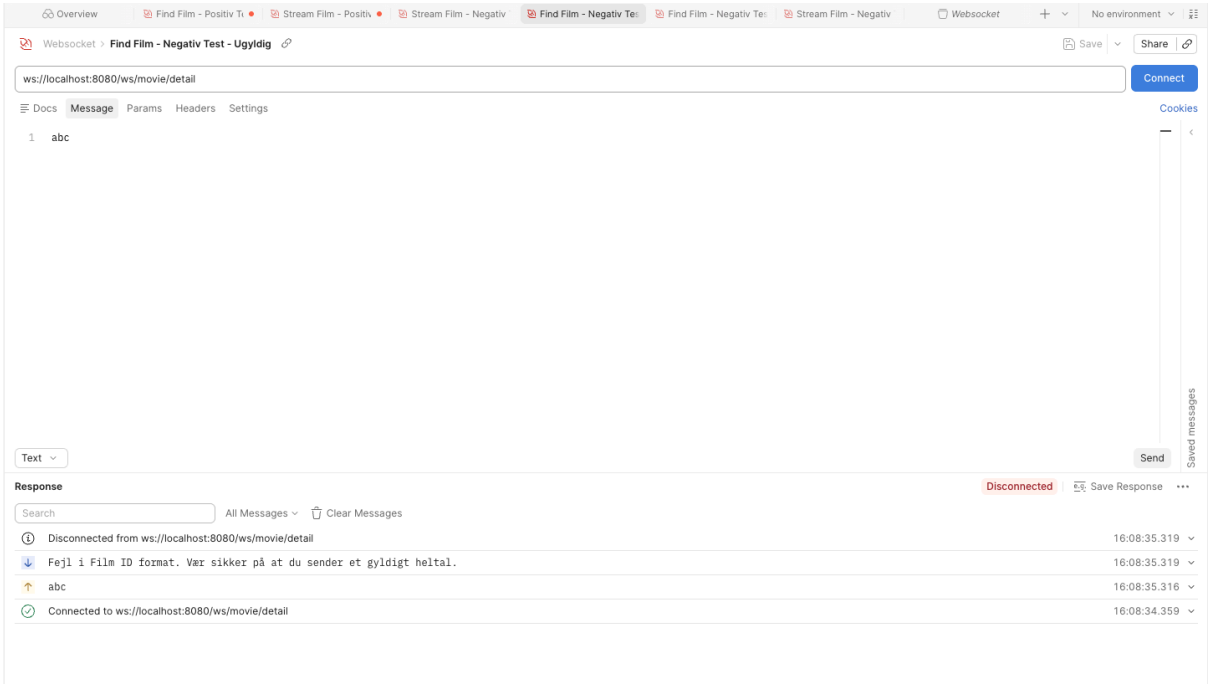
Positive testing: Unary operations

Test 1 - FindMovie



- A valid **movie_id: 1** was sent to the server. And then the server successfully established the Websocket connection and returned a response containing the full movie details for inception.
- This demonstrates that the Unary operation correctly fetches data from the SQLite database and returns the result as JSON through the WebSocket connection.

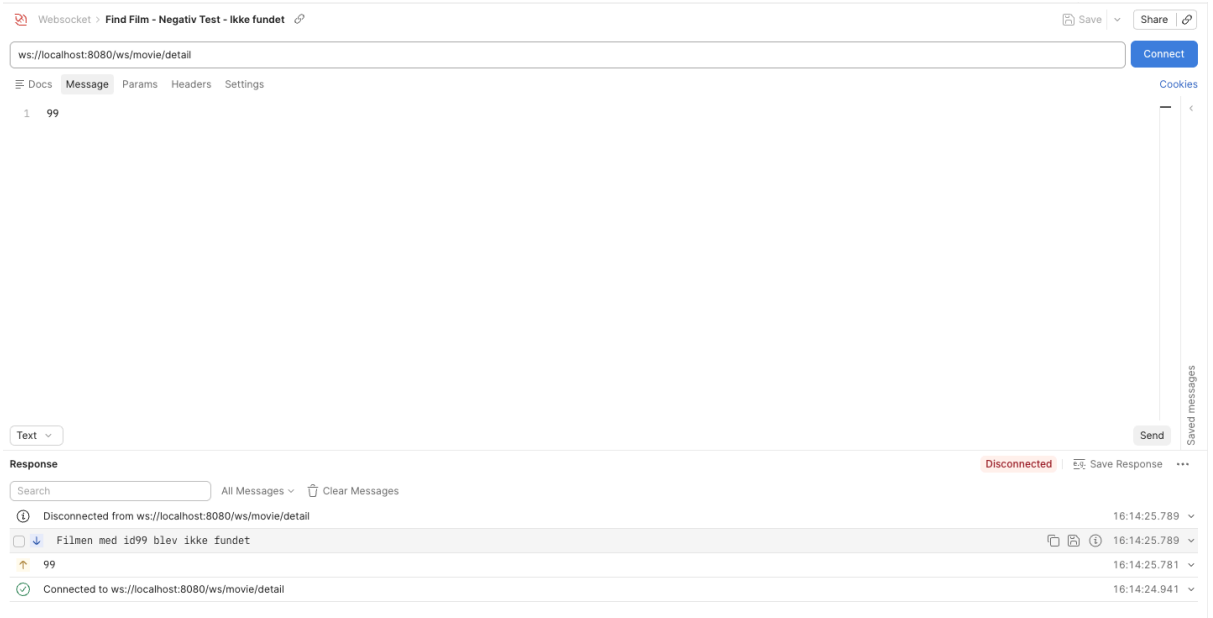
Negative testing: Unary operations



Test 2 - FindMovie (Invalid Characters)

- Sent a invalid character **movie_id: abc** to the server instead of a valid integer value. And the server responded with: "*"Fejl i Film ID format. Vær sikker på at du sender et gyldigt heltal."*

- This demonstrates that the WebSocket service correctly validates the input before processing the request and handles invalid data gracefully by returning a custom error message to the client.



Test 3 - FindMovie (Not Found)

- Sent a valid character **movie_id: 99** that does not exist in the database to the server.
- The server then responded with a custom error message indicating that the movie could not be found: *"Fejl i Film ID format. Vær sikker på at du sender et gyldigt heltal."*
- This demonstrates that the WebSocket service correctly handles requests for non-existing resources and gracefully returns an appropriate error response to the client.

Positive testing: Bidirectional Streaming operations

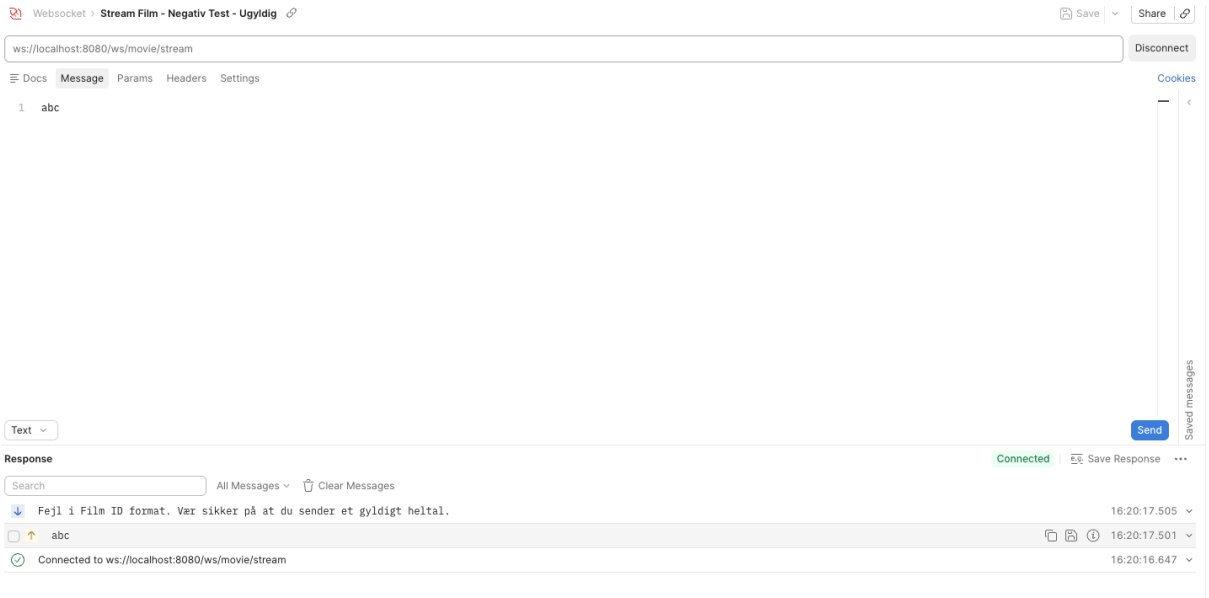
Test 4 - FindMovie

- A valid request: **1** was sent to the server. The server successfully established the WebSocket connection and continuously streamed movie data back to the client every 2 seconds.

- The response contained multiple movie objects from the SQLite database returned as JSON through the WebSocket connection.

- This demonstrates that the Bidirectional Streaming operation correctly keeps the connection open and continuously streams data between the server and client in real time.

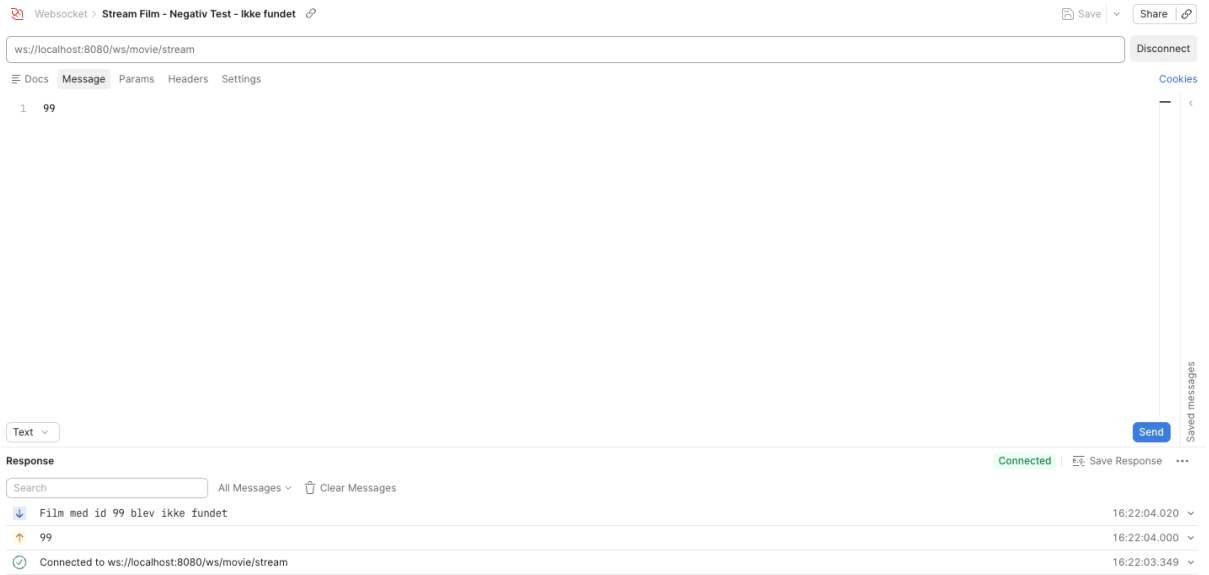
Negative testing: Bidirectional Streaming operations



Test 5 - FindMovie (Invalid Characters)

- Sent invalid characters: **abc** to the server instead of a valid integer value. The server responded with a custom error message indicating that the request format was invalid: *“Fejl i Film ID format. Vær sikker på at du sender et gyldigt heltal.”*

- This demonstrates that the WebSocket service correctly validates the input before processing the streaming request and gracefully handles invalid data by returning an appropriate error message to the client.



Test 6 - FindMovie (Not Found)

- Sent a valid movie_id: 99 that does not exist in the database to the server. The server then responded with a custom error message indicating that the requested movie could not be found: “*Film med id 99 blev ikke fundet”*

- This demonstrates that the WebSocket service correctly handles requests for non-existing resources during streaming operations and gracefully returns an appropriate error response to the client.